ADITYA ANAND

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Indian Institute of Technology Bombay, Mumbai, Maharashtra - 400076, India
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Research Interests

My research interests are in compilers, programming languages, and program analysis. I am interested in designing precise and efficient static and dynamic program analysis techniques for just-in-time (JIT) compilers. My research focuses on combining static analysis with runtime information to enable precise and efficient optimizations in managed runtimes. In particular, my doctoral research developed a robust static+dynamic framework for optimizing object allocation in Java on a production Java Virtual Machine (JVM). I am currently focused on productizing the outcomes of my PhD research with IBM and exploring their application to production compiler and runtime systems.

Work Experience

Indian Institute of Technology Bombay — *Mumbai, India*
Project Associate, Computer Science and Engineering — *Sept 2026 – Ongoing*

Education

Indian Institute of Technology Bombay — *Mumbai, India*
Doctor of Philosophy (Ph.D.), Computer Science and Engineering — *Jul 2023 – Aug 2026*

- **Thesis Title:** *Precision without Regret: Sound and Efficient Staged Analysis for Managed Runtimes*
- **Advisor:** Prof. Manas Thakur
- CGPA: 9.40 / 10.00

Indian Institute of Technology Mandi — *Mandi, India*
Doctor of Philosophy (Ph.D.), Computer Science and Engineering — *Sep 2020 – Jun 2023*
(Transfer to IIT Bombay)

Visvesvaraya Technological University — *Belagavi, India*
Bachelor of Engineering (B.E.), Computer Science and Engineering — *Aug 2015 – Jun 2019*

- CGPA: 9.52 / 10.00
- Department Rank: 4^{th} Rank out of 145 students.

Publications

[1] Gauravsingh Sisodia, Poorna Teja, **Aditya Anand**, and Manas Thakur. **Beyond Java 6: Reviving Reflection-Aware Static Analysis for Modern JVMs**. In *Proceedings of the 18th ACM SIGPLAN International Workshop on Virtual Machines and Intermediate Languages, VMIL'26, p. 8, ACM, 2026.* DOI: 10.1145/3763149

[2] **Aditya Anand**, Vijay Sundaresan, Daryl Maier, and Manas Thakur. **CoSSJIT: Combining Static Analysis and Speculation in JIT Compilers**. In *Proceedings of the ACM on Programming Languages (OOPSLA)*, Singapore, October 2025. DOI: 10.1145/3763149

[3] **Aditya Anand** and Manas Thakur. **Partial Program Analysis for Staged Compilation Systems**. *Formal Methods in System Design (FMSD)*, Springer, June 2024. DOI: 10.1007/s10703-024-00458-x

[4] **Aditya Anand**, Solai Adithya, Swapnil Rustagi, Priyam Seth, Vijay Sundaresan, Daryl Maier, V. Krishna Nandivada, and Manas Thakur. **Optimistic Stack Allocation and Dynamic Heapification for Managed Runtimes**. In *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, Copenhagen, Denmark, June 2024. DOI: 10.1145/3656389

[5] **Aditya Anand** and Manas Thakur. **Principles of Staged Static+Dynamic Partial Analysis**. In *Proceedings of the* 29th *Static Analysis Symposium (SAS)*, Auckland, New Zealand, December 2022. DOI: 10.1145/3563768.3563957

Invited to appear in Formal Methods in System Design (FMSD)

Posters

[P1] **Aditya Anand** et al. **CoSSJIT: Combining Static Analysis and Speculation in JIT Compilers**. Academic Research and Careers for Students (ARCS), ACM India, IIT Hyderabad, February 2026.

[P2] **Aditya Anand** et al. **CoSSJIT: Combining Static Analysis and Speculation in JIT Compilers**. ACM SIGPLAN International Conference on Systems, Programming, Languages, and Applications (OOPSLA/SPLASH Poster Track), Singapore, October 2025.

[P3] **Aditya Anand** et al. **Optimistic Stack Allocation and Dynamic Heapification for Managed Runtimes**. Academic Research and Careers for Students (ARCS), ACM India, Coimbatore, India, February 2025.

[P4] **Aditya Anand**. **A Study of the Impact of Callbacks in Staged Static+Dynamic Partial Analysis**. In *Companion Proceedings of ACM SIGPLAN International Conference on Systems, Programming, Languages, and Applications (SPLASH Companion, Student Research Competition)*, Auckland, New Zealand, December 2022.

Honours and Awards

- **Excellence in Teaching Assistantship Award**, IIT Bombay — Fall 2025
- **George B. Fernandes Award** for Excellence in Ph.D. Research Progress, IIT Bombay — 2025
- **ACM India / IARCS International Travel Grant** — 2025
- **Excellence in Teaching Assistantship Award**, IIT Bombay — Spring 2024
- **ACM India / IARCS International Travel Grant** — 2024
- **ACM SIGPLAN Fellowship**, (PLMW @ PLDI) — 2024
- **Best Poster Presentation Winner**, RISC 24 IIT Bombay — 2024
- **Excellence in Teaching Assistantship Award**, IIT Bombay — Fall 2023
- **GATE Fellowship**, Ministry of Education, Government of India — 2020
- 4th **Rank Holder in B.E. (Computer Science)**, VTU Karnataka — 2019
- 1st **Place** in Inter-College Technical Quiz Competition — 2018
- **Rajya Puraskar** in Bharat Scouts and Guides — 2013

Professional Service

Artifact Evaluation Committee Member:

- Programming Language Design and Implementation (PLDI 2026).
- Object-Oriented Programming, Systems, Languages & Applications (OOPSLA 2026)
- European Conference on Object-Oriented Programming (ECOOP 2025)
- Programming Language Design and Implementation (PLDI 2024).
- European Conference on Object-Oriented Programming (ECOOP 2024)

TEACHINGEXPERIENCE

- **Teaching Assistant**, CS6004: Code Optimization for Object-Oriented Languages — Spring 2026
- **Teaching Assistant**, CS339+CS355: Abstractions and Paradigms for Programming — Fall 2025
- **Teaching Assistant**, CS614: Advanced Compilers — Spring 2025
- **Teaching Assistant**, CS339+CS355: Abstractions and Paradigms for Programming — Fall 2024
- **Teaching Assistant**, CS6004: Code Optimization for Object-Oriented Languages — Spring 2024
- **Teaching Assistant**, CS614: Advanced Compilers — Fall 2023
- **Teaching Assistant**, CS515: Advanced Computer Science Practicum — Fall 2022
- **Teaching Assistant**, CS502: Compiler Design — Fall 2022
- **Teaching Assistant**, CS611: Program Analysis — Spring 2022
- **Teaching Assistant**, CS302: Paradigms of Programming — Spring 2021

INVITED TALKS AND RESEARCH PRESENTATIONS

1. **Invited Talk:** *Precision Without Regret: Sound and Efficient Memory Optimization in Managed Runtimes*, IIT Gandhinagar, India, March 2026.
2. **Invited Talk:** *Precision Without Regret: Efficient Optimization Strategies for JIT Compilation*, IIT Hyderabad, India, February 2026.
3. **Research Presentation:** *CoSSJIT: Combining Static Analysis and Speculation in JIT Compilers*, Research Highlights in Programming Languages (RHPL), India, December 2025.
4. **Conference Paper Presentation:** *Combining Static Analysis and Speculation in JIT Compilers*, OOPSLA 2025, Singapore, October 2025.
5. **Research Presentation:** *Combining Static Analysis and Speculations in JIT Compilers*, Innovations in Compiler Technology (IICT), October 2025.
6. **Research Presentation:** *Program Analysis for Managed Runtimes in Presence of Dynamic Features*, Innovations in Compiler Technology (IICT), September 2024.
7. **Research Presentation:** *Optimistic Stack Allocation and Dynamic Heapification for Managed Runtimes*, Software Engineering Research in India (SERI), July 2024
8. **Invited Talk:** *Optimistic Stack Allocation and Dynamic Heapification for Managed Runtimes*, ComputerSysTalks@India, June 2024.
9. **Conference Paper Presentation:** *Optimistic Stack Allocation and Dynamic Heapification for Managed Runtimes*, PLDI 2024, Copenhagen, Denmark, June 2024.
10. **Research Presentation:** *Staged Static+Dynamic Partial Analysis for Java-like languages*, Software Engineering Research in India (SERI), June 2023.
11. **Conference Paper Presentation:** *Principles of Staged Static+Dynamic Partial Analysis*, Static Analysis Symposium (SAS 2022), Auckland, New Zealand, December 2022.

TECHNICAL SKILLS

- **Core Research Frameworks:** Soot (Java Program Analysis), Eclipse OpenJ9 VM, JavaCC, JTB.
- **Programming Languages:** Java, C, C++, Haskell, Scheme.
- **Systems & Systems Scripting:** Bash, Awk, Git, Docker.

Academic References

Prof. Manas Thakur
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Indian Institute of Technology Bombay
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